

Structure, Ionic Conductivity, and Dielectric Properties of Li-Rich Garnet-type $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) and Their Chemical Stability

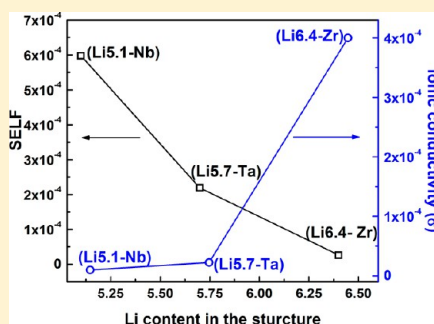
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Supporting Information

ABSTRACT: Lithium garnet oxides are considered as very promising solid electrolyte candidates for all-solid-state lithium ion batteries (SSLiBs). In this work, we present a cubic garnet-type $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) system as a potential electrolyte for SSLiBs. Powder X-ray diffraction (PXRD) and scanning electron microscopy (SEM) were employed to investigate the structural stability of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$. The results from PXRD and SEM suggested structural and morphological transformation as a function of dopant concentration. In addition to Li-ion transport in $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$, the dielectric properties were also investigated in the light of electron energy loss functions, which showed some surface energy loss and negligible volume energy loss for the studied garnets. Surface and volume energy loss functions of a mixed conducting LiCoO_2 was studied for comparison. The long-term chemical stability of one of members, $\text{Li}_{5.3}\text{La}_3\text{Ta}_{1.85}\text{Sm}_{0.15}\text{O}_{12}$, was performed on aged sample using PXRD, SEM, and thermogravimetric analysis.



1. INTRODUCTION

Recently, there has been an increased recall on devices that use Li-ion batteries (LIBs) because of the safety issues associated with the use of organic polymer electrolytes.^{1–4} Whether the cause is short-circuiting or overcharging, the organic electrolyte in LIBs tends to break down and subsequently results in an explosion. Therefore, there is a growing demand of replacing conventional LIBs with safe and robust all-solid-state LIBs (SSLiBs). It is well-known that the conventional electrolytes in Li-ion batteries prominently pose serious limitations, including flammability, limited voltage, solid-electrolyte interphase formation, and contamination impact to the environment in case of improper disposal or recycling.⁵ SSLiBs provide potential solutions to the aforementioned problems associated with using organic polymer electrolytes.

Various researchers have developed solid electrolytes, including lithium nitrides,^{6,7} lithium hydrides,^{8,9} lithium halides,¹⁰ lithium superionic conductors (LISICONs),¹¹ sodium superionic conductors (NASICONs),¹² argyrodites,¹³ perovskites,^{14,15} and garnets.^{16–27} Among them, garnet-type electrolytes have received considerable attention, because they are the most stable interface against Li metal and have a wide electrochemical window (over 6 V/Li).²⁸ An ideal garnet exhibits a general chemical formula of $\text{A}_3\text{B}_2\text{X}_3\text{O}_{12}$ (A = Mg, Ca, Y, La; B = Al, V, Mn, Fe, Ni, Ga, Ge; X = Al, Si, Ge), where A, B, and X are eight, six, and four oxygen-coordinated cation sites.²⁹ A novel family of garnet-like structure electrolytes with Li-rich

oxides $\text{Li}_5\text{La}_3\text{M}_2\text{O}_{12}$ (M = Nb, Ta), initially developed by Thangadurai et al.,³⁰ has very attractive for SSLiBs. Since then, further researches have been performed to understand the physical and chemical properties of these Li-rich garnet-type metal oxides by aliovalent substitution at the La and M sites.^{31–37}

Herein, we report the synthesis, crystal structure, ionic conductivity, dielectric property, and chemical stability of new

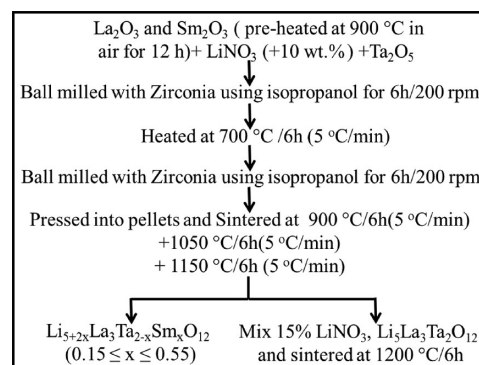


Figure 1. Scheme illustrates the conventional solid-state synthesis method used to prepare Li-stuffed garnet electrolytes with nominal chemical formula of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$).

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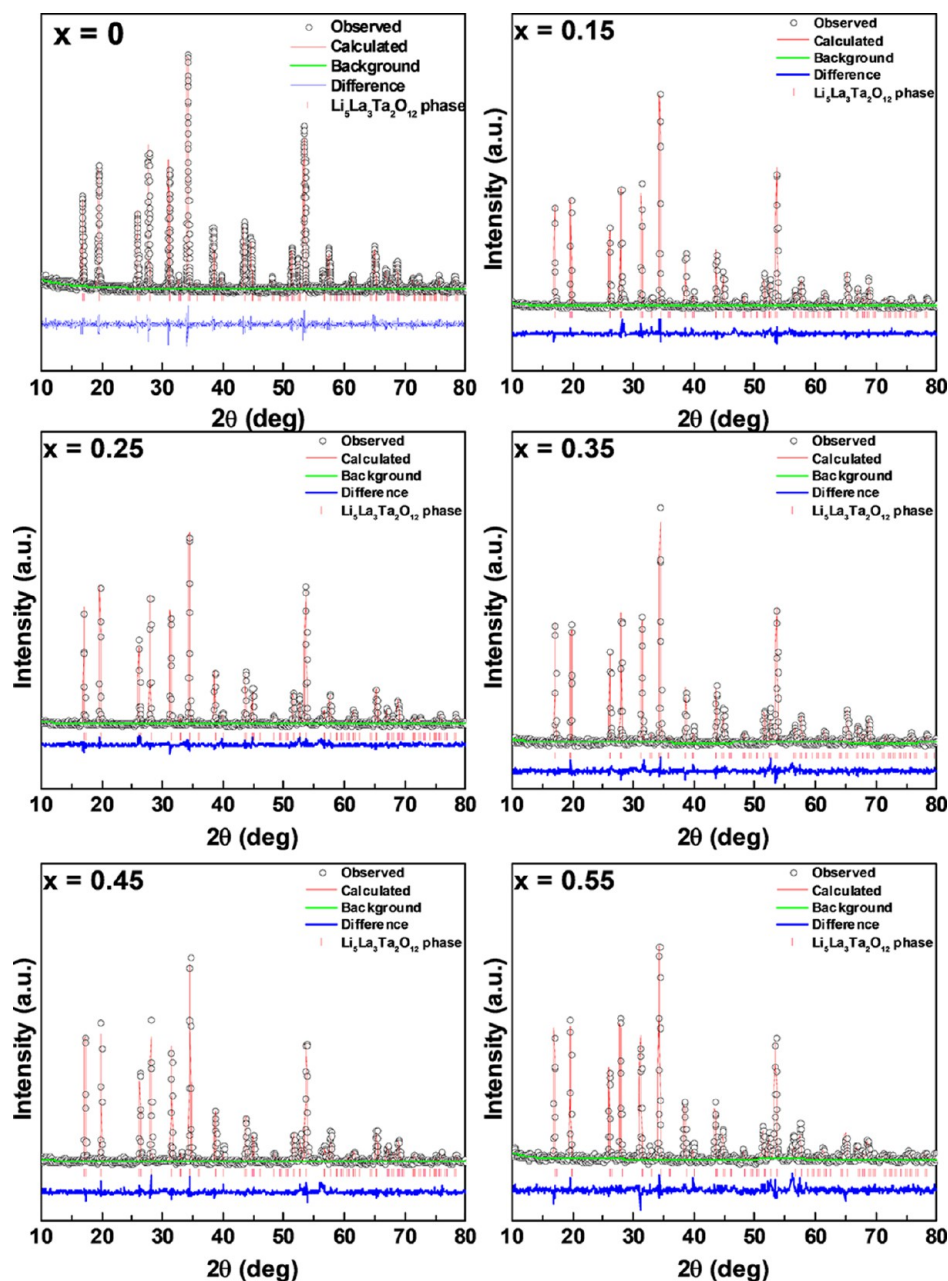


Figure 2. Rietveld refinement results of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) stuffed garnets. All the prepared compounds exhibit the formation of garnet-type structure with space group $Ia\bar{3}d$. To avoid the carbonate formation due to exposure to ambient air, the experiment was performed at a short duration (1 h).

Sm-doped garnet-type $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$). The key motivations behind this study are to introduce parameters such as molar volume (V_m), packing density, compactness (δ), and volume and surface energy loss functions to understand the relations between ionic and electronic transports in Li-rich $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ garnets. The obtained findings could be employed to further improve the ionic conductivity of solid electrolytes for SSLiBs.

2. THEORETICAL CONSIDERATIONS

2.1. Density and Its Related Parameters. Density of solid electrolyte is critical for the application in SSLiBs. Archimedes principle is a common method to determine the experimental density, which depends on impregnating pellets with water or other solvents such as isopropanol. Powder X-ray

diffraction (PXRD) data is also frequently used to estimate the theoretical density of the investigated garnets using the following expression:³⁸

$$\rho_{\text{XRD}} = \frac{MZ}{V_{\text{uc}}N_{\text{A}}} \quad (1)$$

where M is the molecular/formula weight of the sample (g/mol), Z is the number of chemical formula present in a unit cell ($Z = 8$ for the garnet structure), V_{uc} is the volume of the unit cell (cm^3), and N_{A} is Avogadro's number ($6.022 \times 10^{23} \text{ mol}^{-1}$). The theoretical density of garnet samples can also be computed using the relation:³⁹

$$\rho_{\text{atomic}} = \left(\sum_i \frac{m_i}{\rho_i} \right)^{-1} \quad (2)$$

Table 1. Powder X-ray Rietveld Refinement Results for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$)^a

	atom	Wyckoff site	x/a	y/b	z/c	occupancy	U_{iso} ($\text{\AA}^2$)
$x = 0$ R_p : 9.75, $\chi^2 = 1.237$	Li1	24d	1/4	7/8	0	0.802	0.025
	Li2	48g	1/8	0.6826	0.5674	0.139	0.025
	Li3	96h	0.0927	0.684	0.5795	0.147	0.025
	La	24c	1/8	0	1/4	1	0.0296(1)
	Ta	16a	0	0	0	1	0.0268(8)
	O	96h	0.2879(9)	0.0984(1)	0.2011(2)	1	0.0192(9)
$x = 0.15$ R_p : 10.11, $\chi^2 = 1.49$	Li1	24d	1/4	7/8	0	0.718	0.025
	Li2	48g	1/8	0.6826	0.5674	0.181	0.025
	Li3	96h	0.0927	0.684	0.5795	0.173	0.025
	La	24c	1/8	0	1/4	1	0.028
	Ta	16a	0	0	0	0.925	0.024
	Sm	16a	0	0	0	0.075	0.025
$x = 0.25$ R_p : 10.50; $\chi^2 = 1.35$	O	96h	0.2867(5)	0.0979(3)	0.2010(2)	1	0.025
	Li1	24d	1/4	7/8	0	0.681	0.025
	Li2	48g	1/8	0.6826	0.5674	0.215	0.025
	Li3	96h	0.0927	0.684	0.5795	0.182	0.025
	La	24c	1/8	0	1/4	1	0.0330(2)
	Ta	16a	0	0	0	0.875	0.0233
$x = 0.35$ R_p : 10.61; $\chi^2 = 1.50$	Sm	16a	0	0	0	0.125	0.025
	O	96h	0.2835(2)	0.1062(9)	0.1993(1)	1	0.025
	Li1	24d	1/4	7/8	0	0.678	0.025
	Li2	48g	1/8	0.6826	0.5674	0.227	0.025
	Li3	96h	0.0927	0.684	0.5795	0.193	0.025
	La	24c	1/8	0	1/4	1	0.0128(7)
$x = 0.45$ R_p : 10.50; $\chi^2 = 1.47$	Ta	16a	0	0	0	0.825	0.0055(1)
	Sm	16a	0	0	0	0.175	0.025
	O	96h	0.2857(4)	0.0987(5)	0.1988(4)	1	0.025
	Li1	24d	1/4	7/8	0	0.676	0.025
	Li2	48g	1/8	0.6826	0.5674	0.228	0.025
	Li3	96h	0.0927	0.684	0.5795	0.210	0.025
$x = 0.55$ R_p : 10.88; $\chi^2 = 1.62$	La	24c	1/8	0	1/4	1	0.0182(6)
	Ta	16a	0	0	0	0.775	0.01050
	Sm	16a	0	0	0	0.225	0.025
	O	96h	0.2821(3)	0.1052(3)	0.1987(1)	1	0.0596
	Li1	24d	1/4	7/8	0	0.671	0.025
	Li2	48g	1/8	0.6826	0.5674	0.229	0.025
	Li3	96h	0.0927	0.684	0.5795	0.224	0.025
	La	24c	1/8	0	1/4	1	0.0240(4)
	Ta	16a	0	0	0	0.725	0.0170(4)
	Sm	16a	0	0	0	0.275	0.025
	O	96h	0.2857(2)	0.1037(2)	0.1987(6)	1	0.0526(6)

^aThe model used for the Li distribution is based on refs 57,58.Table 2. A Summary of Lattice Constant, Crystallite Size, Experimental, Theoretical and Relative Densities, Molar Volume, Packing Density, and Compactness of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) Garnets

	$\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$					
	$x = 0$	$x = 0.15$	$x = 0.25$	$x = 0.35$	$x = 0.45$	$x = 0.55$
lattice constant ($\text{\AA}$)	12.8176(4)	12.8108(9)	12.8206(9)	12.8309(2)	12.8463(9)	12.8525(1)
crystallite size (nm)	48.6	52.2	47.1	44.3	44.2	37.2
ρ_{exp} ($\text{g}\cdot\text{cm}^{-3}$)	5.04	3.75	3.72	5.60	4.13	4.13
ρ_{XRD} ($\text{g}\cdot\text{cm}^{-3}$)	6.34	6.28	6.27	6.25	6.19	6.19
relative density (%)	79.5	59.7	59.3	89.6	66.7	66.7
ρ_{atomic} ($\text{g}\cdot\text{cm}^{-3}$)	6.26	6.04	5.91	5.78	5.65	5.53
molar volume V_m cm^3/mole	160.55	165.89	169.44	172.97	176.49	179.99
packing density $P \times 10^{22}$ (atom/cm^3)	8.26	8.11	8.00	7.91	7.82	7.73
compactness δ	0.560	0.542	0.53	0.519	0.509	0.499

where m_i is the mass atomic fraction, and ρ_i is the density of the structural unit (e.g., Li, La, Ta, Sm, and O). The theoretical densities were utilized to calculate the packing density P , which

is defined as the ratio of the used space to allocated space via the formula $P = (N_A \rho_{\text{atomic}})/M$. The molar volume (V_m) of the investigated Li-stuffed garnet-like structure is governed by the

atomic mass and the structure of the components. For any composition, V_m can be expressed as $V_m = (\sum_i C_i A_i) / \rho_{\text{atomic}}$ ⁴⁰

where C_i is the atomic fraction of the component i , and A_i is the atomic weight. However, the change in the mean atomic volume V_m due to chemical interactions of the elements forming the network of a given solid could be measured by the compactness (δ), which can be expressed as:⁴¹

$$\delta = \frac{\sum_i \frac{C_i A_i}{\rho_i} - \sum_i \frac{C_i A_i}{\rho_{\text{atomic}}}}{\sum_i \frac{C_i A_i}{\rho_{\text{atomic}}}} \quad (3)$$

The theoretical density from eq 2, packing density, molar volume, and compactness values of $\text{Li}_{5.3}\text{La}_3\text{Ta}_{1.85}\text{Sm}_{0.15}\text{O}_{12}$ were calculated as an example, and given in the Supporting Information.

2.2. Permittivity and Its Related Parameters. Electrochemical impedance spectroscopy (EIS) is a powerful technique for testing and diagnosing solid electrolyte.⁴² Particularly, electrical conductivity as a function of frequency provides information on ion dynamics in solid electrolytes. The bulk conductivity (σ) could be determined from complex impedance spectra data using the formula $\sigma = d/AR_b$, where R_b is the garnet's bulk resistance, d is the pellet thickness, and A is surface area of the electrode. Arrhenius plot of the bulk conductivity could be extracted from the impedance spectra. Dielectric relaxation process and ionic conductivity could be estimated using permittivity spectra. The dielectric constants of the investigated garnets were determined with the use of the real and imaginary parts of compositions' impedance, that is:⁴³

$$\epsilon' = \frac{Z''}{\omega C_0(Z'^2 + Z''^2)} \quad (4)$$

$$\epsilon'' = -\frac{Z'}{\omega C_0(Z'^2 + Z''^2)} \quad (5)$$

where ϵ' and ϵ'' are real and imaginary parts of the dielectric constants, and $C_0 = \epsilon_0(A/d)$ is the vacuum capacitance of the cell, where ϵ_0 is the permittivity of free space ($8.854 \times 10^{-14} \text{ F cm}^{-1}$). The inelastic electron scattering in the solid can be ascribed due to energy loss function.^{44,45} It has the advantage of covering the complete energy range including valence interband transitions and core level excitations.⁴⁵ The most favored forms of the energy loss functions are volume energy loss function (VELF) and surface energy loss function (SELF), i.e.,^{46,47}

$$\text{VELF} = \frac{\epsilon''}{\epsilon'^2 - \epsilon''^2} \quad (6)$$

$$\text{SELF} = \frac{\epsilon''}{(\epsilon' + 1) + \epsilon''^2} \quad (7)$$

Both VELF and SELF represent the rate of energy loss due to electrons while passing through the bulk material and on its surface, respectively.^{44,46,48,49} These two functions show the effect of electronic conductivity on the garnet structure, which has an impact on understanding the total conductivity of the studied garnets.

3. EXPERIMENTAL ASPECTS

Samples of nominal chemical composition $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ were prepared via solid-state synthesis method, the same as described in our previous work,⁵⁰ and the synthesis procedure is also summarized in Figure 1. The phase of the garnet powders was characterized by PXRD using Cu K α radiation (40 kV, 40 mA; Bruker AXS D8) in the angular

range of $2\theta = 10^\circ - 80^\circ$ with a step scan of $0.02^\circ \text{ min}^{-1}$. The Rietveld refinements were implemented by the general structure analysis system (GSAS) program⁵¹ and the EXPGUI interface⁵² to extract the lattice parameters and relative phase fractions. The experimental density of the cylindrical pellets was determined using Archimedes principle by immersing the pellets in isopropanol for half an hour prior to the weighing step.

The morphology of the garnet samples' pellets was examined by scanning electron microscopy (SEM; Zeiss Sigma VP), where the grain size of the pellets was determined as well. Thermogravimetric analysis (TGA; Mettler Toledo Thermal Analysis, TGA/DSC1/1600HT) with a temperature range of 25–1100 °C at a rate of $10^\circ \text{ C min}^{-1}$ in air was performed on the garnet powder samples. Solartron electrochemical impedance spectroscopy gain-phase analyzer (SI model no. 1260) was used to measure samples' impedance (circular pellets $\sim 10 \text{ mm}$ diameter and 2 mm thickness) in the frequency range of 0.1 Hz–1 MHz at 25–300 °C. Au paste (Heraeus Inc., LP A88–11S, Germany) was painted on both sides of the pellets and cured at 600 °C for 1 h to eliminate any additional effect due to organic binders. Further, an experimental setup combining a gastight quartz tube and a symmetrical cell was gradually heated and maintained at the desired temperature for 2 h prior to the impedance measurements. For further alternating current (AC) measurements, purchased lithium cobalt oxide (LiCoO_2 , 97%, Alfa Aesar) was pressed into pellets (circular pellets $\sim 12 \text{ mm}$ diameter and 1.5 mm thickness) for 4 min without any binders. The pellets were heated at 900 °C for 24 h. Subsequently, Au paste (Heraeus Inc., LP A88–11S, Germany) was painted on both sides of the pellets and dried at 400 °C for 1 h.

4. RESULTS AND DISCUSSION

4.1. Structural Analysis and Microstructure Studies.

Rietveld refinement results of six stuffed garnet compositions in

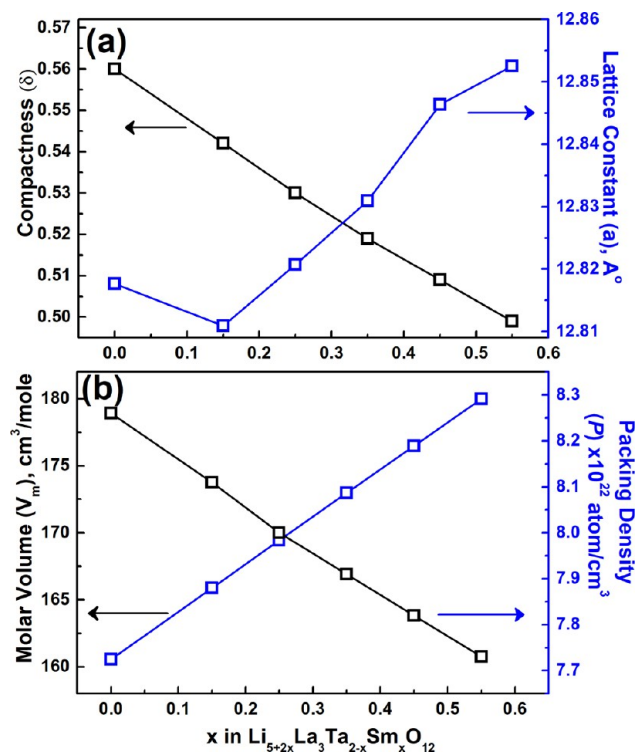


Figure 3. (a) Compactness (δ) (eq 3) and lattice constant (a) obtained from the Rietveld refinement values for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) Li-stuffed garnet type as a function of the doping concentration. (b) Molar volume calculated from $V_m = (\sum_i C_i A_i) / \rho_{\text{atomic}}$ and packing density (P) calculated from $P = (N_A \rho_{\text{atomic}}) / M$ values for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) Li-stuffed garnet type as a function of the doping concentration.

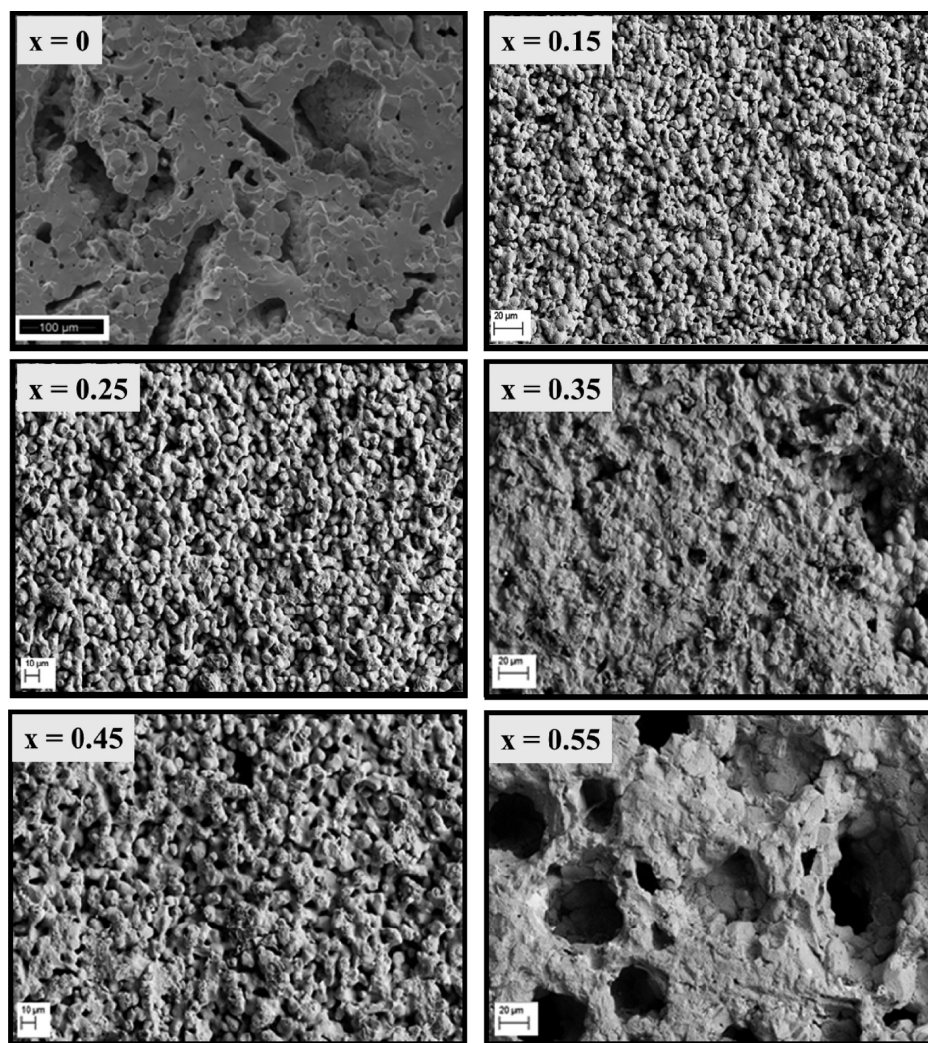


Figure 4. Typical SEM images show the surface morphology of as-prepared Li-stuffed garnet type $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$).

$\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) are shown in Figure 2. In general, garnet electrolytes could exhibit two crystal symmetries namely, cubic or tetragonal.^{53–55} The PXRD pattern of $x = 0$ member, $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$, matches well with that in the literature,⁵⁶ suggesting a cubic phase. For all members of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$, the characteristic peaks of Sm-doped garnets retain the cubic phase without any major secondary phases. The result of PXRD Rietveld refinement data (Table 1) exhibits the Li^+ ion distribution in $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ over octahedral 48g/96h and tetrahedral 24d sites. Furthermore, the refinement results identified all phases of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) in a cubic structure with space group of $Ia\bar{3}d$. However, octahedral 48g/96h site is less favorable than tetrahedral site 24d for all samples.^{57,58} So, Li^+ is accommodated more in the tetrahedral site 24d for the studied structures. Also, it is clear that Sm^{3+} mainly substitutes Ta^{5+} at the octahedral 16a site. The lattice constant parameters for all garnet compositions were evaluated from the structure refinement and summarized in Table 2. As anticipated, solid solution follows Vegard's Law, where the lattice parameter increases linearly with increasing cation substitution (Table 2). This change in the lattice parameter is well-predicted, since the sixfold coordination ($\text{Ta}^{5+} = 0.64 \text{ \AA}$) was replaced by higher ionic radius with same coordination atoms ($\text{Sm}^{3+} = 0.958 \text{ \AA}$).⁵⁹ This aspect could be further confirmed theoretically by

Hume–Rothery rule, which states that extensive substitution solid solution occurs only if the relative difference between the atomic radii of the two species is less than or equal to 15%.⁶⁰ $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ has two available sites (La and Ta) for Sm substitution. However, opposite trend could have been anticipated if Sm substitutes in the eightfold coordinated La site because La ($\text{La}^{3+} = 1.16 \text{ \AA}$) has higher ionic radius than Sm ($\text{Sm}^{3+} = 1.079 \text{ \AA}$).

To further understand the physical properties of garnets, crystallite size, experimental, theoretical and relative densities, molar volume, packing density, and compactness of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ were estimated and listed in Table 2. The experimental density for all samples was found to be lower than their corresponding theoretical density. This observation can be referred to the defect formation in the crystals that reduces the total mass per unit volume.⁶¹ The nominal garnet $\text{Li}_{5.7}\text{La}_3\text{Ta}_{1.65}\text{Sm}_{0.35}\text{O}_{12}$ exhibited the highest experimental density (5.60 g cm^{-3}). This nature of $x = 0.35$ member appears later to coincide with Li-ion conductivity of this composition. The trend is also in agreement with the conclusion reported earlier in our previous work⁵⁰ and by Ni et al.,⁶² where the high bulk Li-ion conductivity of Li-stuffed garnets was found to be associated with the highest density values. Furthermore, all of density, packing density, and compactness values tend to decrease gradually, as Sm and Li

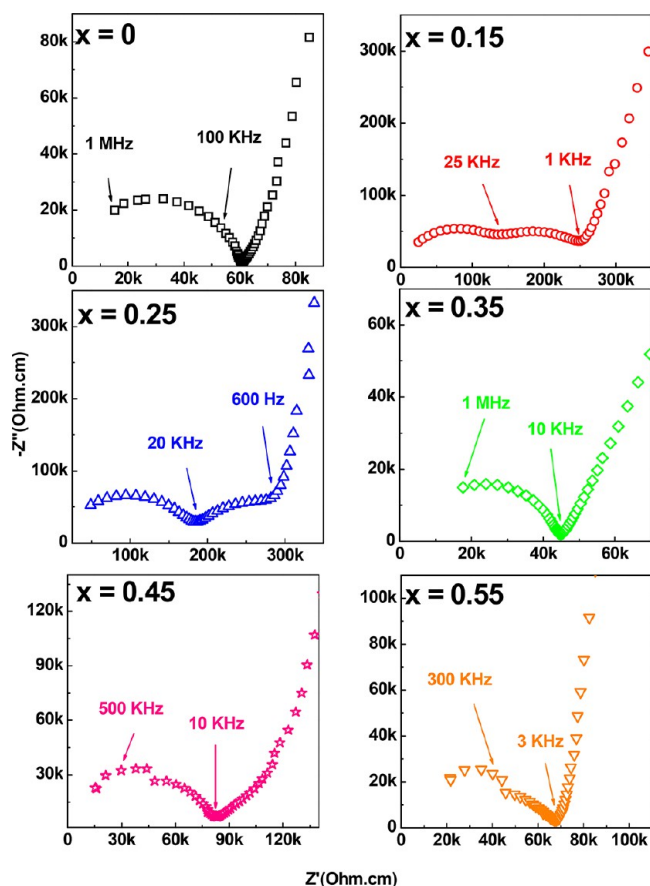


Figure 5. Nyquist (Cole–Cole) plots of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) Li-stuffed garnet type at room temperature in ambient air.

contents in $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ are increased. The reduced packing density might be due to the large atomic radius as well as mass of Sm compared to that of Ta, while the decrease of the compactness values with the increase of Sm content confirms the increase in compositions' free volume. The values of compactness and lattice constant as a function of the doping concentration are also shown in Figure 3a. The molar volume shows an opposite behavior to the density. The molar volume (V_m) and packing density (P) values for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ electrolytes as a function of the doping concentration are shown in Figure 3b. Overall decrease in the crystallite size is observed with increasing the doping concentration. On the basis of these observations, controlling the molar volume and consequently the density of the garnets is related to both the atomic fraction and the atomic mass of the elements forming the garnet structure. Admittedly, knowing this concept could improve tailoring new garnets with higher density and consequently higher ionic conductivity values. However, other parameters such as heating rate, sintering temperature, and time should be taken under considerations to improve the density.⁶³

SEM study was utilized to investigate the surface morphology of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ sintered pellets (Figure 4). From the SEM images, we observed that the particles are uniformly distributed. The SEM images of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ confirm that the particle size and distribution mainly depend upon the extrinsic doping concentration. Clearly, the increase of the Sm and Li contents has a high impact on the morphology of the samples. When the concentration of Sm and Li contents is increased, the porosity of the structure was found to be increased

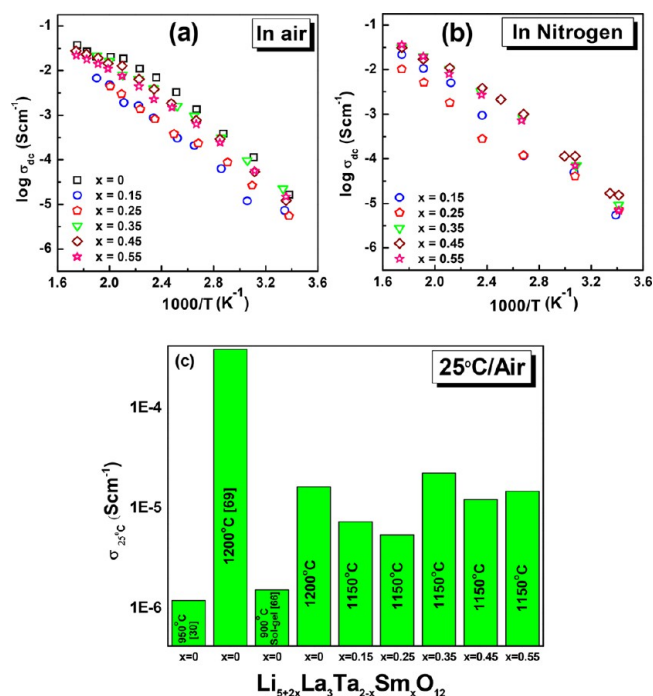


Figure 6. Arrhenius plots for DC conductivity of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) garnet electrolytes in (a) air and (b) nitrogen. (c) Ionic conductivity of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) garnet electrolytes and comparison with reported ionic conductivity of the parent phase $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ in literature^{30,56,68} at 25 °C in air. Each ionic conductivity value was labeled by the final sintering temperature used to prepare the electrolyte.

(Table 2). As illustrated in Figure 4, the highest doped $\text{Li}_{6.1}\text{La}_3\text{Ta}_{1.45}\text{Sm}_{0.55}\text{O}_{12}$ has pores ranging from 10 to 30 μm . For this reason, the extrinsic doping in $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ hinders the particles from better connectivity. The spherical shape particles with the size in micron range can also be observed for all compositions. In addition, parent phase $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ sintered at 1200 °C forms agglomerates in the range of 5–10 μm and exhibits a mixed inter- and trans-granular fracture mode.

4.2. Electrochemical Impedance Measurements. EIS was performed as a function of extrinsic doping, temperature, and atmosphere to distinguish contributions from the bulk and the grain-boundary over the frequency range (0.1 Hz–1 MHz). The ionic conductivity of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ was examined in air and nitrogen using Au electrodes. A typical representative Nyquist plot of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$, measured at room temperature in air, is displayed in Figure 5. The main characteristics of the impedance spectrum are a resistance offset at high frequencies, which reflects the bulk properties of a material, a compressed semicircle at intermediate frequencies, and a tail at low frequencies due to electrode effect.^{64,65} The higher-frequency intercept and the compressed semicircle can be attributed to bulk and grain-boundary contributions for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($x = 0.15, 0.25$). However, only bulk contribution has been observed for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($x = 0, 0.35, 0.45, 0.55$). The bulk conductivity increases with increasing Sm content until $x = 0.35$ and then decreases. This observation strongly agrees with the overall decrease in the density values, which decreased with increasing Sm content in $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ (Table 2). Notably, the ionic conductivity of $2.25 \times 10^{-5} \text{ S cm}^{-1}$ is determined for

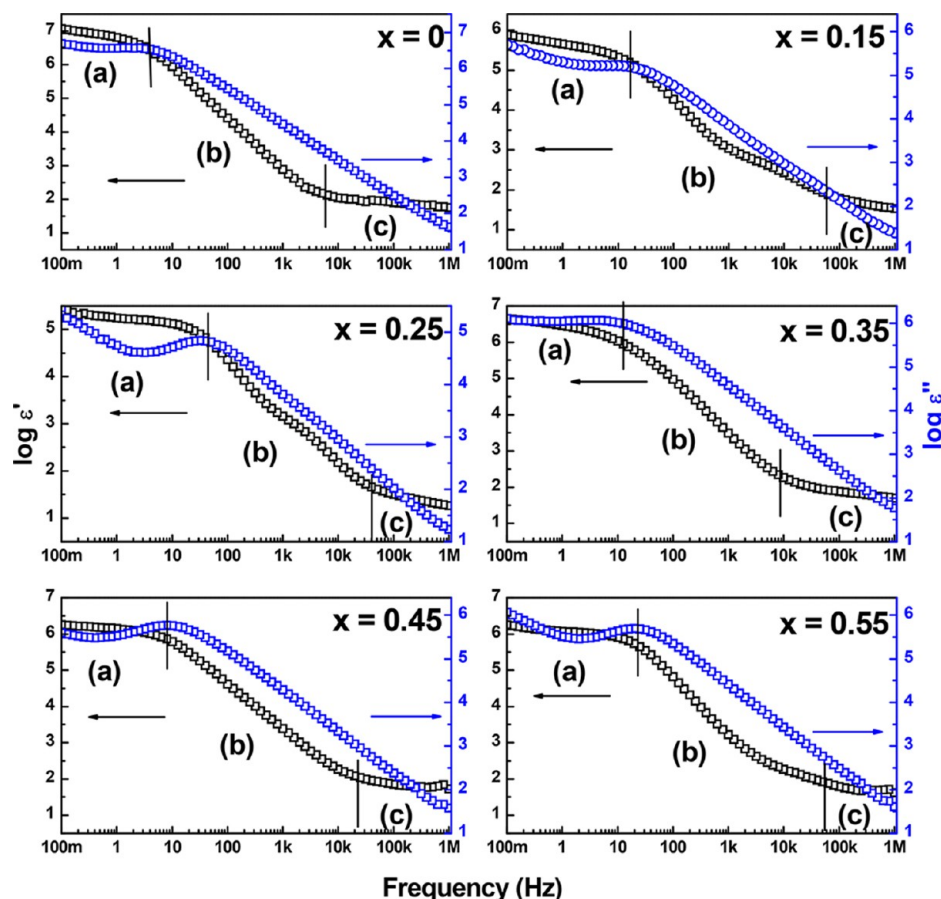


Figure 7. Real and imaginary parts of permittivity as a function of frequency for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) Li-stuffed garnet type at room temperature in air. Regions (a–c) represent the low, intermediate, and high frequencies ranges' contributions to the real dielectric constant, respectively.

$\text{Li}_{5.7}\text{La}_3\text{Ta}_{1.65}\text{Sm}_{0.35}\text{O}_{12}$ at ambient condition to be highest among all compositions studied in this study. Moreover, the ionic conductivity of $\text{Li}_{5.7}\text{La}_3\text{Ta}_{1.65}\text{Sm}_{0.35}\text{O}_{12}$ was the same in N_2 and air at room temperature. The impedance plots for all samples as a function of temperature is shown in Figure S1. The bulk conductivity (σ) for the Sm-doped garnets shows Arrhenius behavior⁶⁶ in air and N_2 (Figure 6a,b). In general, the solid-state ionic conductivity has a close relationship with grain size; the material with large grains and fewer grain boundaries usually has higher conductivity.⁶⁷ Also, the Li-ion conductivity values of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ garnet electrolytes with activation energy ranging from 0.39 to 0.42 eV were compared with the reported ionic conductivity of the parent phase $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ in literature^{30,56,68} at 25 °C in air (Figure 6c). The highest ionic conductivity of $3.9 \times 10^{-4} \text{ S cm}^{-1}$ was shown by $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ prepared at 1200 °C/24 h.⁵⁶

4.3. Dielectric Constants and Related Parameters. To further elucidate the electrical properties of the investigated Sm-doped $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ garnets, the dielectric constants were computed using the impedance data. The real and imaginary permittivities as a function of frequency at room temperature are shown in Figure 7. At low frequencies, a plateau region (a) is obtained for ϵ' and ϵ'' in all plots, which could be interpreted to the electrode–electrolyte interface polarization. The mobile Li^+ ions migrate over large distance and accumulate near the electrode at lower frequencies, which form layers of space charge that block the electric field. This process enhances the electrical polarization and has high ϵ' and ϵ'' values at low

frequencies region (a). At intermediate frequencies, region (b), a broad dispersion is present in which the dielectric constants decrease with increasing frequency. This trend is a typical behavior for ion-conducting materials.⁶⁹ In this region, ions wait for the surrounding ions to reorganize themselves. However, at high frequencies, region (c), a short plateau region is observed for ϵ' but ϵ'' continues to decrease. In this region, electrical polarization does not occur, since the charge displacement may be small compared to the dimension of the conducting region.⁵⁸ Real and imaginary permittivity spectra over the temperature range as a function of frequency $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ are given in Supporting Information (Figures S2 and S3). An overall increase of in ϵ' and ϵ'' values is observed with increasing temperature, which reveals that the dipolar polarization is a thermally activated process.

Interestingly, complex dielectric constant, $\epsilon^* = \epsilon' + i\epsilon''$,^{70–72} is important in terms of extracting dielectric loss tangent or dissipation factor ($\tan \delta$), the volume energy loss function (VELF), and surface energy loss function (SELF). Figure 8 represents the variation of the dielectric loss factor $\tan \delta = \epsilon''/\epsilon'$ ⁷³ as a function of frequency for $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ garnets at 25–300 °C. The $\tan \delta$ shows a relaxation peak due to bulk ionic transport in the temperature range of 25–150 °C for all investigated compositions, except $x = 0.15$, and 0.25 members of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$. These two members exhibit the presence of two distinct relaxation peaks at two resonant frequencies, one in the low-frequency and another at high-frequency region due to ionic transport process of the

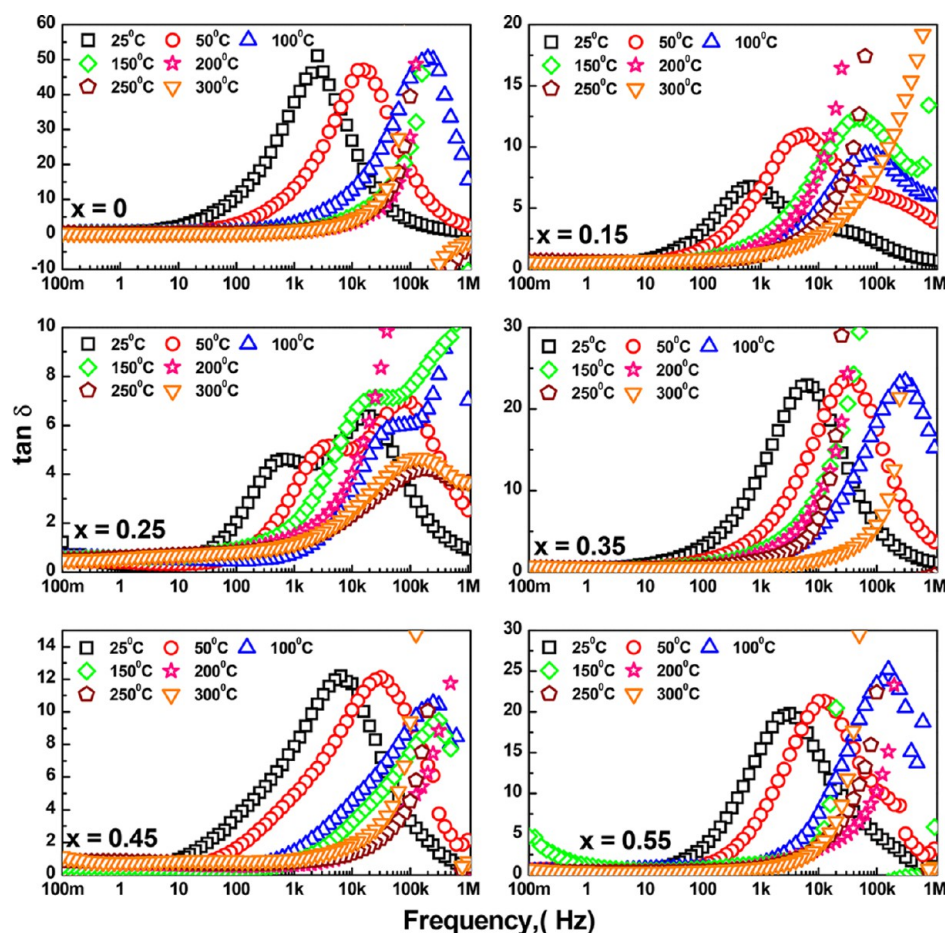


Figure 8. Dielectric loss ($\tan \delta$) as a function of frequency of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) Li-stuffed garnet type at different temperatures.

grain-boundary and bulk, respectively. Furthermore, all observed relaxation peaks show an almost increase in the peak intensity and shift to higher frequency region with increase in temperature at the same time.

For comparison, dielectric properties of predominately electronic conducting LiCoO_2 cathode were also studied, and Nyquist plot is shown in Figure S4. The dependence of SELF and VELF on energy for LiCoO_2 and $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ samples at 25 °C is illustrated in Figure 9. As expected, highly electronic conducting LiCoO_2 exhibits higher SELF-value comparing with $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ garnets. Both Li-stuffed garnets $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ and LiCoO_2 show higher SELF values than the VELF. It is observed that a clear trend is raised in the case of SELF; however, a pattern with specific peaks at high frequency range is obtained in the case of VELF at room temperature. Those peaks match with the bulk and grain-boundary range in the complex impedance plot. Moreover, the steady state in VELF pattern matches with the low-frequency electrode effect range of impedance. For SELF plot the noticeable change in the energy loss values starts at high-frequency range and neither surface nor volume energy loss in the range of electrode effect. The $x = 0$ and 0.35 members reveal nearly the lowest SELF. This could be the reason for having the highest conductivity in these two compositions as compared to other investigated garnets. The $x = 0.15$ and 0.25 members exhibit the highest SELF, and they showed grain-boundary contribution to the ionic conductivity. This concept could be explained as the fact that a grain-boundary is a crystallographic mismatch zone with possibly positive charge in which the

electron accumulated and consequently increases the SELF values.⁷⁴ Figures S5 and S6 show SELF and VELF of Li-stuffed garnets $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ as a function of the energy at different temperatures, respectively. Figure S7 shows SELF and VELF of LiCoO_2 as a function of the energy at different temperatures. It is clearly seen that increasing temperature is accompanied by the decrease in both SELF and VELF for all compositions including LiCoO_2 . The highest loss of energy (SELF and VELF) is noticed at lower temperatures (25–50 °C). At higher temperatures (100–300 °C), all the investigated compositions showed nearly the same extent of energy loss. Li-stuffed garnets exhibit higher SELF and lower VELF, which is opposite to other materials such as chalcogenides.⁷² This reflects that the Li-stuffed garnets insulate electrons except small portion, which is critical for the potential candidate for electrolytes in batteries and other energy storage devices.

To monitor these parameters, SELF and VELF for other Li-stuffed garnets were calculated using the dietetic constants values extracted from the literature.⁷⁵ Figure S8 shows SELF and VELF as a function of energy for $\text{Li}_{5.5}\text{La}_3\text{Nb}_{1.75}\text{Y}_{0.2}\text{O}_{12}$, $\text{Li}_6\text{La}_3\text{Nb}_{1.5}\text{Y}_{0.5}\text{O}_{12}$ and $\text{Li}_{6.5}\text{La}_3\text{Nb}_{1.25}\text{Y}_{0.75}\text{O}_{12}$ at –22 °C. It has been reported that $\text{Li}_{6.5}\text{La}_3\text{Nb}_{1.25}\text{Y}_{0.75}\text{O}_{12}$ composition has the highest ionic conductivity of $1.27 \times 10^{-5} \text{ S cm}^{-1}$ and $2.99 \times 10^{-4} \text{ S cm}^{-1}$ at –22 °C and 25 °C, respectively. This coincides with its lowest SELF values compared with ($\text{Li}_{5.5}\text{-Nb}$) and ($\text{Li}_6\text{-Nb}$) phases. To further elucidate, Li contents impact on the SELF and ionic conductivity was studied. Figure 10 shows a comparison between the observed ionic conductivity and SELF for $\text{Li}_{5.7}\text{La}_3\text{Ta}_{1.65}\text{Sm}_{0.35}\text{O}_{12}$ in our study and other Li-stuffed

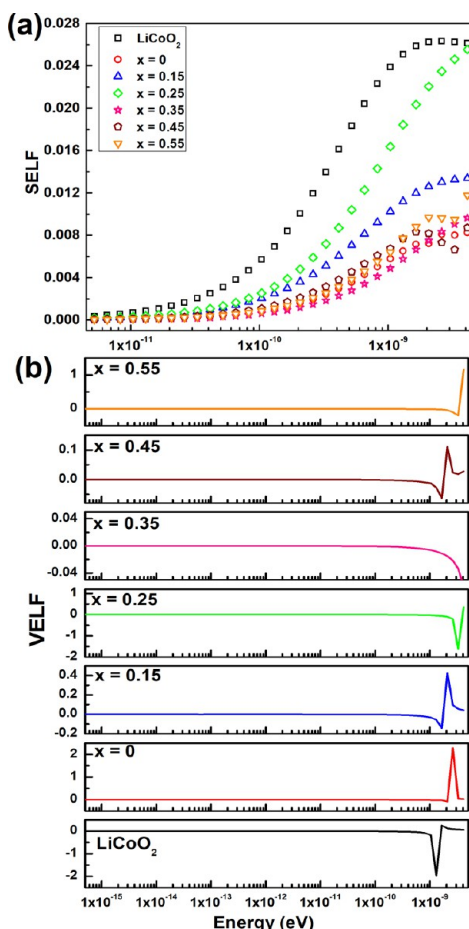


Figure 9. Plot of the variations of (a) the SELF and (b) the VELF with the energy for mixed electronic ionic conductor LiCoO_2 and Li-stuffed garnet type $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$) samples at 25°C under study.

garnets $\text{Li}_{5.1}\text{La}_3\text{Nb}_{1.95}\text{Gd}_{0.05}\text{O}_{12}$ ⁷⁶ and $\text{Li}_{6.4}\text{Ga}_{0.2}\text{La}_3\text{Zr}_2\text{O}_{12}$.⁷⁷ As ionic conductivity and Li content increase, the value of SELF decreases, and this is consistent with aforementioned observations (Figure 9). In general, the trend of SELF and VELF seems to be the properties related to the garnet materials.

4.4. Structural Stability Study. To further explain the chemical stability of garnets in CO_2 , the pellets of $\text{Li}_{5.3}\text{La}_3\text{Ta}_{1.85}\text{Sm}_{0.15}\text{O}_{12}$ were exposed to air for 2 days, thermally etched at $900^\circ\text{C}/6\text{ h}$, and chemically etched using $0.5\text{ M HNO}_3/15\text{ min}$. Figure 11 shows cross-section SEM images for air-exposed, thermally etched, acid-etched $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ pellets. Clearly, the sample exposed for 2 days in air shows Li_2CO_3 layer on the surface and surrounding the grains. This behavior could be explained as there are two kinds of sites to accommodate Li ions in the cubic phase as explained earlier, the tetrahedral sites 24d and the octahedral sites 48g/96h sites. It is well-known that Li ions partially occupy both sites with leftover vacant sites to enable the hopping of the mobile ions. However, 24d sites are more favorable than 48g/96h, and this is the case in $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$. On one hand, the unequal distributions of Li ions results in unrelaxed structure that tends to stabilize by forming Li_2CO_3 layer between the unstable Li ions in the structure and CO_2 from air. Moreover, Li_2CO_3 layer formation could be removed totally by acid etching. On the other hand, it is partially removed by using thermal etching (Figure 11). This could be because of

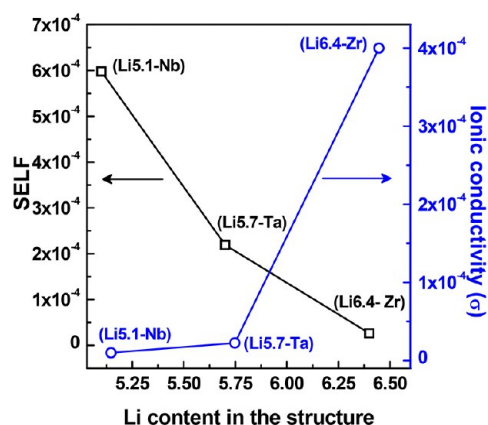


Figure 10. Variations of the SELF and the ionic conductivity with the Li content in the structure for $\text{Li}_{5.1}\text{La}_3\text{Nb}_{1.95}\text{Gd}_{0.05}\text{O}_{12}$ ($\text{Li}_{5.1}\text{-Nb}$)⁷⁶ $\text{Li}_{5.7}\text{La}_3\text{Ta}_{1.65}\text{Sm}_{0.35}\text{O}_{12}$ ($\text{Li}_{5.7}\text{-Ta}$) and $c\text{-Li}_{6.4}\text{Ga}_{0.2}\text{La}_3\text{Zr}_2\text{O}_{12}$ ($\text{Li}_{6.4}\text{-Zr}$).⁷⁷ The SELF values were calculated at the same frequency data point that gives the ionic conductivity value for each composition.

carbonate reformation (CO_2 at $\leq 600^\circ\text{C}$) during the cooling process. SEM images indicate that there is no change in the structure after the etching treatments; however, only Li_2CO_3 as secondary phase was removed. Nevertheless, further studies are needed to firmly establish the influence of Li_2CO_3 layer formation and etching treatments on garnet's ionic conductivity and Li content and distribution in the structure.⁷⁸

With these preliminary results on the chemical stability, PXRD studies were also performed on $\text{Li}_{5.3}\text{La}_3\text{Ta}_{1.85}\text{Sm}_{0.15}\text{O}_{12}$ after seven months of its synthesis to determine the positions of Li_2CO_3 accommodation inside the structure. Rietveld refinement for the aged $\text{Li}_{5.3}\text{La}_3\text{Ta}_{1.85}\text{Sm}_{0.15}\text{O}_{12}$ sample is shown in Figure 12, and the corresponding data are listed in Table 3. As shown in Figure 12 and Table 3, in addition to the 95.05% $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ garnet phase, 2.58% lithium carbonate (Li_2CO_3) phase was detected as secondary phase, which accommodate 8f and 4e Wyckoff sites with space group $\text{C}12/c1$. Also, 1.41% secondary phase of $\text{Sm}(\text{OH})_3$ was detected at 2c and 6h Wyckoff sites with space group $\text{P}63/m$. The secondary phases might have been introduced due to the tendency of garnets to exchange lithium with protons in ambient air.^{79–82} The structural model used to refine the structure of aged sample was based in the literature.^{57,58,83–85}

TGA of the aged samples was also analyzed to clarify the chemical stability of Sm-doped samples. TGA of $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ is shown in Figure S9. All curves feature weight loss events in the range from ca. $300\text{--}800^\circ\text{C}$, indicating the same nature of interaction with air for all samples but different in the extent of interaction. Approximately less than 1% weight loss was observed for $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$, which agrees with its specific synthesis technique. However, 7% weight loss was detected for $\text{Li}_{5.3}\text{La}_3\text{Ta}_{1.85}\text{Sm}_{0.15}\text{O}_{12}$. Many weight loss steps were observed, which are related to H_2O and CO_2 releasing.

5. CONCLUSION

A pure cubic phase with a space group $\text{Ia}\bar{3}\text{d}$ has been observed for garnet-type $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ($0 \leq x \leq 0.55$). An overall increase in the lattice parameter and molar volume values have been observed for $0 \leq x \leq 0.55$ with the substitution of pentavalent Ta ions for the trivalent Sm ions. Density, packing density, and compactness values show

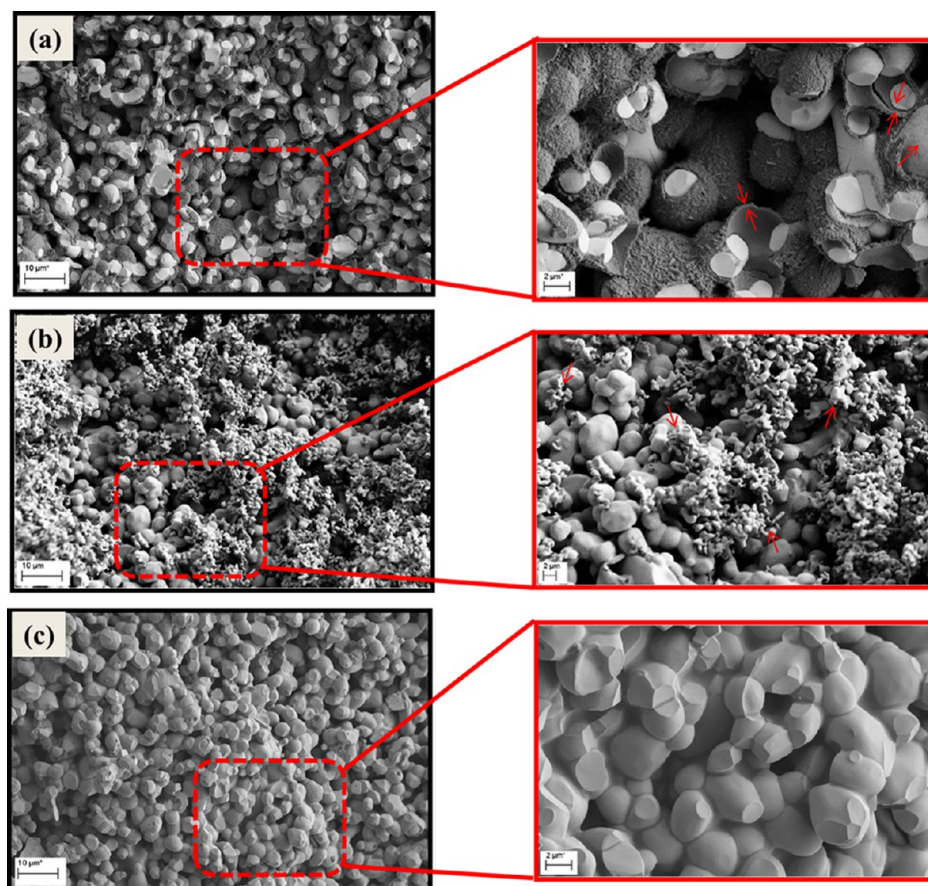


Figure 11. SEM images show the cross-section morphology of (a) 2 days air-exposing, (b) thermal-etched, and (c) 0.5 M HNO₃ etched (15 min) Li_{5+2x}La₃Ta_{2-x}Sm_xO₁₂ electrolyte, respectively. Higher magnification images on the right-hand side are given for better illustration.

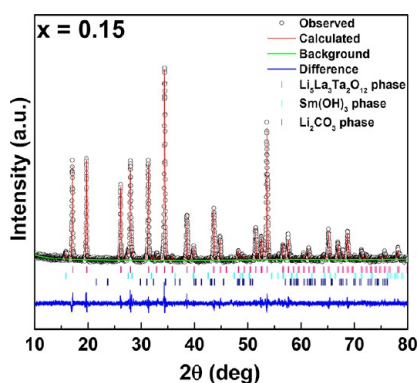


Figure 12. Rietveld refinement results of Li_{5.3}La₃Ta_{1.85}Sm_{0.15}O₁₂ aged sample. In addition to the primary Li₅La₃Ta₂O₁₂ garnet phase, secondary phases of Li₂CO₃ and Sm(OH)₃ are also detected.

decreasing behavior with increasing Sm and Li contents in Li_{5+2x}La₃Ta_{2-x}Sm_xO₁₂. SEM images indicate that with increasing the concentration of Li and Sm contents, the porosity of the structure increases, and the density was found to be decreased. The highest Sm-doped Li_{6.1}La₃Ta_{1.45}Sm_{0.55}O₁₂ has pores ranging from 10 to 30 μm. Li_{5.7}La₃Ta_{1.65}Sm_{0.35}O₁₂ exhibited the 2.25×10^{-5} S cm⁻¹ ionic conductivity, which is the highest among all studied Sm samples. Dielectric properties study has been utilized to introduce new parameters such as the volume (VELF) and surface energy loss functions (SELF). Dielectric properties study has been utilized to introduce the volume (VELF) and surface energy loss functions (SELF) as new

parameters. Garnet-type compositions and LiCoO₂ exhibit SELF higher than VELF. The air stability study reveals that no change in the structure of aged Li_{5.3}La₃Ta_{1.85}Sm_{0.15}O₁₂ sample after the etching treatments; however, only a secondary phase of Li₂CO₃ was removed. Rietveld refinement for the aged Li_{5.3}La₃Ta_{1.85}Sm_{0.15}O₁₂ showed two secondary phases' formation, namely, lithium carbonate Li₂CO₃ and samarium hydroxide Sm(OH)₃. Many weight-loss steps have been observed for the TGA of the aged Li_{5+2x}La₃Ta_{2-x}Sm_xO₁₂ ($0 \leq x \leq 0.55$) samples, which are related to H₂O and CO₂ releasing.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acs.inorgchem.7b00816](https://doi.org/10.1021/acs.inorgchem.7b00816).

AC Impedance analysis and TGA of Li_{5+2x}La₃Ta_{2-x}Sm_xO₁₂ ($0 \leq x \leq 0.55$) (PDF)

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Author Contributions

D.A. synthesis, experiments, data analysis, and manuscript writing; S.M. PXRD analysis and Rietveld refinement; D.A, M.E. and S.F. dielectric, SELF, and VELF calculations, data

Table 3. Powder X-ray Rietveld Refinement Result for the Aged Sample ($x = 0.15$) in $\text{Li}_{5+2x}\text{La}_3\text{Ta}_{2-x}\text{Sm}_x\text{O}_{12}$ ^a

atom	Wyckoff site	x/a	y/b	z/c	occupancy	U_{iso} ($\text{\AA}^2$)
95.05% phase 1: $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ ($a = 12.8458(6)$ $\text{\AA}$; space group = $Ia\bar{3}d$)						
Li1	24d	1/4	7/8	0	0.715	0.025
Li2	48g	1/8	0.6826	0.5674	0.179	0.025
Li3	96h	0.0927	0.684	0.5795	0.174	0.025
La	24c	1/8	0	1/4	1	0.0236(6)
Ta	16a	0	0	0	0.925	0.0303
Sm	16a	0	0	0	0.075	0.025
O	96h	0.2871(8)	0.1034(4)	0.1936(7)	1	0.025
1.41% phase 2: $\text{Sm}(\text{OH})_3$ ($a = 6.5223(8)$ $\text{\AA}$, $c = 3.8491(9)$ $\text{\AA}$; space group = $P63/m$)						
atom	Wyckoff site	x/a	y/b	z/c	occupancy	U_{iso} ($\text{\AA}^2$)
Sm1	2c	1/3	2/3	1/4	1	0.025
O1	6h	0.2975(2)	0.2832(7)	1/4	1	0.025
H1	6h	0.3938	0.3113	1/4	1	0.025
2.58% phase 3: Li_2CO_3 ($a = 8.3465(4)$ $\text{\AA}$, $b = 4.9739(2)$ $\text{\AA}$, $c = 6.1447(1)$ $\text{\AA}$; space group = $C12/c1$)						
atom	Wyckoff site	x/a	y/b	z/c	occupancy	U_{iso} ($\text{\AA}^2$)
Li1	8f	0.1968	0.4454	0.8334	1	0.025
C1	4e	0	1/3	1/4	1	0.025
O1	4e	0	1/3	1/4	1	0.025
O2	8f	0.1334(3)	0.0230(4)	0.2880(1)	1	0.025

^a $R_p = 9.36$; $\chi^2 = 1.32$.

analysis, and manuscript writing; V.T. experimental design, data analysis, and manuscript writing; all authors reviewed the manuscript.

Notes

The authors declare no competing financial interest.

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